

3D Shape Build Challenge

Planrs Academy

Learn, Build, and Have Fun with 3D Shapes!

Let's explore Spheres, Cubes, Cuboids, Cones and Cylinders!

■ Subject	■ Level	■ Duration
Mathematics – Geometry	Early Primary (Grades 1–2)	45–60 minutes

1. Learning Objectives

By the end of this lesson, students will be able to:

Objective 1	Recognise and name 5 common 3D shapes: Sphere, Cube, Cuboid, Cone, and Cylinder.
Objective 2	Describe each shape using key vocabulary: faces, edges, corners, curved, flat, rolls, stacks.
Objective 3	Match each shape to familiar real-world Indian objects found at home and in daily life.

2. Lesson Opener: The Shape Hunt in Ramesh's Kitchen

Ramesh from Chennai is helping set the breakfast table. He notices interesting shapes all around him! The round orange on the table, his steel tumbler for chai, and the rectangular biscuit box all have different shapes. This opening scene invites students to go on a *shape hunt* in their own everyday surroundings.

Round Orange: A perfectly round orange sitting on the table – no corners anywhere! → Sphere

Steel Tumbler: A shiny steel tumbler for chai – round at top and bottom! → Cylinder

Biscuit Box: A rectangular biscuit box with flat sides and corners! → Cuboid

3. The Five 3D Shapes

1

Sphere

Perfectly Round with No Flat Parts!

◆ Properties

- Curved surface only
- Rolls in any direction
- No edges, no corners

◆ Indian Examples

- Laddoo
- Cricket ball
- Orange

◆ More Examples

- Marble
- Coconut
- Tennis ball

2

Cube

A Box Shape with All Sides as Perfect Squares!

◆ Properties

- 6 flat square faces
- 12 edges
- 8 corners – all faces equal

◆ Indian Examples

- Dice (pasa)
- Sugar cubes
- Ice cube

◆ More Examples

- Rubik's Cube
- Small gift box
- Chalk piece

3

Cuboid

A Box Shape Like Cube But Stretched!

◆ Properties

- 6 flat faces (rect./sq.)
- 12 edges
- 8 corners

◆ Indian Examples

- Tiffin box (dabba)
- Biscuit packet
- Diwali sweet box

◆ More Examples

- Brick
- Mobile phone
- Book

4

Cone

Round Flat Bottom with a Pointed Tip at the Top!

◆ Properties

- 1 flat circular face (base)
- 1 curved surface
- 1 pointed vertex (apex)

◆ Indian Examples

- Ice-cream cone
- Party hat (topi)
- Traffic cone

◆ More Examples

- Shankha tip
- Cone-shaped leaf cup
- Funnel

5

Cylinder

Two Round Flat Ends and Curved Surface in the Middle!

◆ Properties

- 2 flat circular faces
- 1 curved surface
- Can roll and stack

◆ Indian Examples

- Dholak / Tabla drum
- Steel tumbler
- Pressure cooker

◆ More Examples

- Bournvita tin
- Bangle tube box
- Battery

4. Shape Comparison Table

Use this table to compare all five shapes across key properties.

Shape	Flat Faces	Curved Surface	Edges	Corners	Rolls?	Stacks?
Sphere	0	Yes	0	0	✓ Yes	✗ No
Cube	6	No	12	8	✗ No	✓ Yes
Cuboid	6	No	12	8	✗ No	✓ Yes
Cone	1	Yes	0	1	✓ Yes	✗ No
Cylinder	2	Yes	0	0	✓ Yes	✓ Yes

✓ Rolling shapes: Sphere, Cone (on its side), Cylinder. ✓ Stacking shapes: Cube, Cuboid, Cylinder.

5. Shape Detective Game

Look at each object below and say which 3D shape it is. Answers are given to help teachers/parents guide the session.

Round Objects

Cricket ball	→ Sphere – curved surface, rolls everywhere, no corners.
Orange	→ Sphere – perfectly round, no edges.

Box Shapes

Parle-G biscuit packet	→ Cuboid – 6 rectangular faces, 12 edges, 8 corners.
Ludo dice (pasa)	→ Cube – 6 equal square faces.
Tiffin box (dabba)	→ Cuboid – faces are rectangles.

Other Shapes

Steel water bottle	→ Cylinder – 2 circular ends, curved side.
Birthday party hat (topi)	→ Cone – circular base, pointed apex.
Dholak drum	→ Cylinder – 2 round faces, curved surface.

6. Activity: Build Your Sweet Shop!

Students use shape models (or household objects) to build a colourful **mithai (sweet) shop** display. This hands-on activity consolidates shape recognition, stacking, and spatial thinking.

1

Step 1 – Build the Shelves

Stack cuboid and cube boxes on top of each other to create shelves for the shop. Discuss: How many faces touch when you stack? Why do spheres not stack?

2

Step 2 – Arrange the Sweets

Place sphere-shaped laddoos on the top shelf. Arrange cone-shaped leaf cups (or ice-cream cones) in a row along the front. Count the faces, edges, and corners of each shape.

3

Step 3 – Complete the Display

Place a cylinder (dholak drum, tin, or tumbler) at the side of the shop as a decorative element. Students label each object with its shape name on a sticky note.

7. Shape Song ■ Sing Along and Learn!

Use this rhyme to help students remember all five shapes. Clap on each beat or tap the desk!

Sphere is round like my laddoo sweet,

Cube has six square faces neat,

Cuboid is like my tiffin box,

Cone points up like party hat top,

Cylinder rolls like my dholak drum —

Five 3D shapes, we know every one!

8. Key Vocabulary

Use these terms when describing and comparing 3D shapes.

Face	A flat surface of a 3D shape. A cube has 6 faces.
Edge	A line where two faces of a shape meet. A cube has 12 edges.

Corner / Vertex	The point where edges meet. A cube has 8 corners.
Flat	A smooth, level surface with no curve (like the face of a cube).
Curved	A rounded surface with no flat parts (like the surface of a sphere).
Rolls	A shape moves by turning when pushed. Spheres, cylinders, and cones roll.
Stacks	A shape can be placed on top of another. Cubes and cuboids stack easily.
Apex	The pointed tip of a cone – also called its vertex.

9. Take-Home Activity: Shape Hunt at Home!

Students draw the objects they find in their notebook and share with the class the next day. Happy shape hunting, little explorers!

Shape(s) to Find	What to Look For	Draw & Label
Sphere & Cube	Find something ROUND with no flat face (Sphere). Find a box where ALL faces are the same size (Cube).	■
Cuboid & Cone	Find a box with DIFFERENT-sized faces (Cuboid). Find something with a POINT at the top (Cone).	■
Cylinder	Find something with TWO round flat ends, like a tin, tumbler, or bottle (Cylinder).	■

10. Teacher / Parent Notes

★ **Bring real objects:** Before the lesson, collect a laddoo (or ball), a dice, a tiffin box, an ice-cream cone, and a steel tumbler to demonstrate each shape physically.

★ **Encourage Tamil vocabulary:** Invite students to share the Tamil names for shapes and objects (e.g., ■■■■■ for sphere, ■■■■■ for cone, ■■■■■ for cylinder) to build bilingual understanding.

★ **Differentiation:** For advanced students: ask how many faces, edges, and corners each shape has and record in a table. For support: focus only on identifying shapes by touch/sight.

★ **Assessment:** Use the Shape Detective Game as a quick formative assessment. Listen to student explanations for evidence of understanding.

★ **Extension activity:** Challenge students to build a 'shape tower' using only stackable shapes and record which ones work best and why.

■ You Are a Shape Champion! ■

Today you learned 5 amazing 3D shapes, how to find them in your world, how to describe and compare them, and how to build with them.

Keep looking for shapes everywhere you go!